19 DEC 2022

Roll No.: ______________________

Amrita Vishwa Vidyapeetham

School of Computing, Bengaluru

B.Tech. Degree Examinations December 2022

SeventhSemester

Computer Science and Engineering

19CSE353 MINING OF MASSIVE DATASETS

Duration: Three hours Maximum: 100 Marks

CO	Course Outcome
CO01	Understand the importance of data to business decisions, strategy and behavior. Predictive analytics, data mining and machine learning as tools give new methods foranalyzing massive data sets.
CO02	Expose the students to Big data systems like Hadoop, Spark and Hive.
CO03	Explore means to deal with huge document databases and infinite streams of data to mining large social networks and web graphs. Also, learn Algorithms suitable for largescale mining.
CO04	Use of analytic tools, case-studies to provide first hand insight into how big-data problems and their solutions are commercially derived.
CO05	Design Large Scale Machine Learning algorithms with practical hands-on experience for analyzing very large amounts of data.

Answer all questions

1. Discuss any five challenges faced by Data Mining.Discuss thefive areas of applications of Data Mining. [5+5] [CO03] [BTL 2]

2. List any eightData Mining techniques.Explain any four with an example. [2+8] [CO01] [BTL 1]

3. i. Calculate the Jaccard Similarity using the following two datasets, E & F.

   E= ['cat', 'dog', 'hippo', 'monkey']
   F = ['monkey', 'rhino', 'ostrich', 'tiger'].

   ii. How do you analyse large graphs? Discuss in detail. [2+1+7] [CO03] [BTL 2]

4. What are Hash functions used for in Data Mining?Discuss how is hash table used to build an index with a neat diagram. Discuss the three essential steps for finding the similar Documents. [2+5+3][CO02][BTL2]

5. Government of India has assigned you the task to count the population of India. You can demand all the resources you want, but you have to do this task in 2 months only. Calculating the population of such a large country is not an easy task for a single person. So what would be your approach? Discussthe relevant Programming Model with a neat block diagram. [2+8] [CO04] [BTL 4]

6. Suppose we have the following dataset having various transactions in the Table-1.Find the frequent itemsets usingApriori algorithm. Discuss the two advantages and disadvantages each of Apriori algorithm.

Transaction ID	Items
T1	Carrot, Cucumber, Onion
T2	Carrot, Cucumber
T3	Carrot, Apple, Banana
T4	Banana, Apple
T5	Carrot, Banana, Apple

Table -1

[6+4] [CO03] [BTL 3]

7. How do you decide the optimal number of K in the K Means Algorithm? Divide the given data set into minimum number of Clustersusing K Means algorithm from the below Table-2. Mention the final Cluster(s).
(Use Euclidean distance metric)

Id	X	Y
1	1	1
2	1.5	2
3	3	4
4	5	7
5	3.5	5
6	4.5	5
7	3.5	4.5

Table – 2

Does centroid initialization affect K Means Algorithm? Justify. [1+8+1] [CO01] [BTL 3]

8. Explain the 4 types of Linkages used in Clustering with necessary equations. Discuss the importance of the Dendrogram in Clustering with an example. [8+2] [CO03] [BTL 2]

9. Justify whether it is possible for Netflix to store an entire stream of a Movie at once. How does Netflix make critical calculations about the stream using a limited amount of (secondary) memory? Discuss the necessary Streaming Model to deal with this situation. [1+4+5] [CO04] [BTL 4]

10. YouTube entice users with relevant suggestions based on the choices they make. Discuss the twotypes of Recommendations that maybe used by YouTube. Discuss the two approaches to Recommender systems with appropriate examples. [5+5] [CO03] [BTL 2]

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Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO01	**20**	BTL 1	**10**
CO02	**10**	BTL 2	**50**
CO03	**50**	BTL 3	**20**
CO04	**20**	BTL 4	**20**
CO05	--	BTL 5	--
CO06	--	BTL 6	--